Challenge Name: Echolocation

Category: REV

Description:

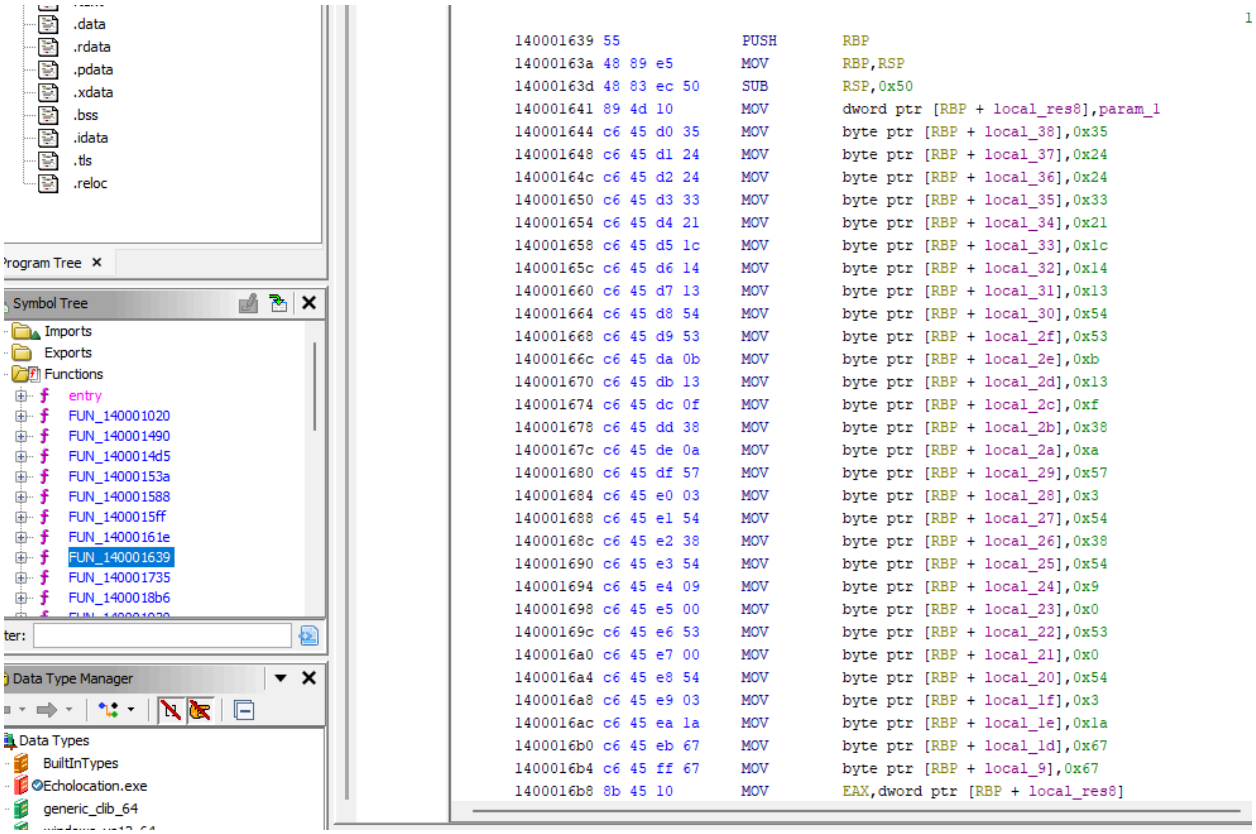
No noise, no trace, just a silent pulse in the dark. The flag is down there. Go listen.

Hint(s):

None

Walkthrough:

There's a good amount of functions in this binary, but looking through the functions briefly, there's one that's really suspicious at 0x140001639 that has a bunch of single byte values in stack variables.



Right beneath that, there's a loop where bytes are XORed by local_9, which is equal to 0x67. The disassembler in Ghidra can help make this clearer, but it's XORing local_9 with all the stack variables we saw before.

```
for (local_10 = 0; local_10 < 0x1c; local_10 = local_10 + 1) {
  local_38[local_10] = local_38[local_10] ^ local_9;
}
```

```
1400016b4 c6 45 ff 67     MOV        byte ptr [RBP + local_9],0x67
1400016b8 8b 45 10        MOV        EAX,dword ptr [RBP + local_res8]
1400016bb 89 c1           MOV        param_1,EAX
1400016bd e8 5c ff        CALL       FUN_14000161e
          ff ff
1400016c2 85 c0           TEST       EAX,EAX
1400016c4 74 03           JZ         LAB_1400016c9
1400016c6 f6 55 ff        NOT        byte ptr [RBP + local_9]

                      LAB_1400016c9                                   XREF
1400016c9 c7 45 f8        MOV        dword ptr [RBP + local_10],0x0
          00 00 00 00
1400016d0 eb 1c           JMP        LAB_1400016ee

                      LAB_1400016d2                                   XREF
1400016d2 8b 45 f8        MOV        EAX,dword ptr [RBP + local_10]
1400016d5 48 98           CDQE
1400016d7 0f b6 44        MOVZX      EAX,byte ptr [RBP + RAX*0x1 + -0x30]
          05 d0
1400016dc 32 45 ff        XOR        AL,byte ptr [RBP + local_9]
1400016df 89 c2           MOV        EDX,EAX
1400016e1 8b 45 f8        MOV        EAX,dword ptr [RBP + local_10]
1400016e4 48 98           CDQE
1400016e6 88 54 05 d0     MOV        byte ptr [RBP + RAX*0x1 + -0x30],DL
1400016ea 83 45 f8 01     ADD        dword ptr [RBP + local_10],0x1

                      LAB_1400016ee                                   XREF
1400016ee 83 7d f8 1b     CMP        dword ptr [RBP + local_10],0x1b
1400016f2 7e de           JLE        LAB 1400016d2
```

XORing all the bytes on the stack reveals the flag RCCTF{st34lth_m0d3_3ng4g3d}

Flag:

RCCTF{st34lth_m0d3_3ng4g3d}